AI Infrastructure Industry – Nvidia, Astera Labs, Broadcom & Intel – Former Senior Executive, Product Marketing at Cadence Design Systems Inc

Consultant | 23 September 2025

Specialist Background

- Over 20 years' experience in the AI infrastructure industry, with strong knowledge of key players such as Nvidia, Astera Labs, Broadcom and Intel
- Well-placed to discuss the AI infrastructure industry's update, the AI Infra Summit, as well as OpenAI's announcement and what it means to the custom ASIC (application-specific integrated circuit) TAM
- Well-versed in Nvidia's dominance in the GPU space and Rubin CPX, a dedicated inference GPU, as well as the Nvidia-Intel partnership

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Transcription begins

Analyst:
I focus on a lot of the AI plays. I think since the last time we chatted, we've had Hot Chips, we've had AI Infra Summit, and then there's been a series of announcements by Nvidia and a few others. Really just hoping to get your updated thought process around some of the dynamics that we've been digging into in the past as it relates to custom vs third-party merchant GPUs and what the scale of networking dynamic might look like and all of the peripheral componentry around there.

On the Intel-Nvidia collaboration, as I think about it from the server side of things, it's the first foray into enabling x86 with NVLink via NV Fusion. Previously, there was some cynicism around the likelihood that hyperscalers really would care too much about NV Fusion, largely to do with the fact that to have their own custom ASIC [application-specific integrated circuit], whether it's Trainium, TPU [tensor processing unit] or MTIA [Meta Training and Inference Accelerator] or Maia and all those kinds of things that might be working behind the scenes, you'd still need the Arm-based CPU. Does anything change in your mind as it relates to hyperscalers as opposed to enterprise customers looking to adopt this future flavour of rack-scale solution that would come with a custom Intel x86 CPU, and then presumably it would be an Nvidia GPU with NV Fusion?

I don't think much has changed from my perspective as it relates to hyperscaler incentives to go this route. Based on initial industry reactions, whether or not the status holds true that NV Fusion is maybe not something that hyperscalers are considering or if this development might make it more worthwhile to dig into and potentially deploy at the time that those products would become available.

Specialist:
This whole integration of Intel with Nvidia is a game-changer for Intel and even for Nvidia as such because whatever said and done, there is still Intel servers being deployed in the industry. By having this collaboration and putting the RTX GPUs inside the Intel desktops and Intel CPUs, it basically gives a path for Intel CPUs to talk to the Nvidia fabric or the Nvidia mesh through the NVLink. If I'm not mistaken, the RTX GPU does support NVLink to connect both these GPUs together to get a better performance. This also creates a doubt of Intel's plans to make their own GPU. Intel has a programme called as Jaguar Shores which is in public domain. It puts us in a puzzle that if this is the case that they are going with Nvidia today for the CPUs [sic], then what happens to the future of Jaguar Shores? Are they completely giving up on implementing Intel-based GPUs? Intel has been struggling in the GPU space for a couple of few years, if you noticed. They haven't really come up with a very strong message on the GPU side of the story. That is one aspect of it, but this is just the beginning of engagement where Nvidia's dominance in the GPU space is now going beyond the Nvidia domain.

It's proliferating into the x86 domain as well with Intel as the beginning. Then what happens to AMD as a GPU vendor because AMD is talking about UALink, is talking about their MI400 and the future 500. That also puts in a doubt because ultimately, AMD is also on an x86 architecture. It'll be putting an immense pressure on AMD that should they be also collaborating with Nvidia and Nvidia being the dominant player in the industry for GPUs taking over both of these players, Intel and AMD. It's a very strategic move on part of AMD. Intel being Intel and it's struggling financially, it gives them a boost that now with the collaboration with the strongest GPU vendor, they have a path to play in a better or a bigger space where they were not being considered previously. Also, it puts a doubt in the space of hyperscalers that if Intel is flexible of putting Nvidia GPUs inside their CPUs, is Intel open to do custom hyperscaler accelerator inside their CPU? As an example, tomorrow, the likes of Amazon or Google can go to Intel and tell them, "Okay, guys, this is my part of accelerator, which is unique to my workload, which my architects have designed. I'm willing to buy thousands of your CPUs per year, but I want this accelerator to be integrated inside your CPU."

What is the reason why these customers today, the hyperscalers are making their own CPUs? They are making their own CPUs not because they are saving any money out of it, but they're making the CPUs because it's optimised to their workload. It gives them superior performance. Plus, they are optimising it for the power and performance. They don't need some generic CPUs to run over there, which is not optimal for their workload. Either it's underutilised or over-utilised. Adding their accelerator in an Intel CPU gives them the path that they don't have to take the heavy-lifting of manufacturing, designing and doing all the stuff with ARM. If Intel meets their requirements and puts their accelerator inside their CPU, they should be open to it. It's a complete different domain. I think so the Intel management has realised that they cannot keep the doors locked to Intel Inside and they have to open up to the new paradigm in the industry where the CPU is governed by the hyperscalers and the workloads what they are using.

Analyst:
Let's say Amazon were to go to Intel and look for a custom 86 CPU that would be integrated with their Trainium accelerator, I don't think that NVLink enablement would be possible in that instance because it would have to be an Nvidia GPU. Is Intel doing a customised CPU enough to make Amazon, for example, shift its roadmap, or is the main goal to ultimately use their own accelerators? If there's no NVLink enablement in that case, how does that change how they think about when we get to '27 and '28, whether or not they go with something more akin to Graviton plus Trainium plus a non-Nvidia fabric vs now they've got the option to do their CPU with their accelerator and NVLink or an x86 custom CPU with an Nvidia GPU and NVLink? Is NVLink the main thing that is important here, and so I should think that it will be something that goes the NVLink route from a scale-up networking standpoint, or do you still think there's room for things like scale-up Ethernet with Tomahawk Ultra or UALink eventually? How does that impact those initiatives on the networking side of things?

Specialist:
A good question again. The NVLink dominance will be restricted to only the Nvidia side of the GPUs, irrespective to whether it's Amazon or Google or any organisation. How much ever they try, they want to basically limit their dependency on Nvidia GPUs to the NVLink to the bare minimum. They have a whole back-end network, which is the non-Nvidia GPUs because Nvidia GPUs are extremely expensive for these organisations. When they're building their own XPUs, they are exploring today two parts. One is the UALink, and other one is the Ethernet scale-up. Ethernet scale-up has got more legs to run. In the coming few months and I would say one year or so, the Ethernet scale-up will be one of the dominant factors for two reasons. Number one is because already the Ethernet deployment is already existing in this infrastructure. Number two, the dominance of players like Broadcom who are really pushing the envelope to adopt to Ethernet scale-up and not to look at the UALink. UALink by construction is an accurate protocol much better than Ethernet scale-up. It's in line with NVLink. I think so we discussed this before also. It basically truly is on the memory semantics what you need, but the problem with

UALink is the ecosystem. For some reason, AMD lost the focus or they did not put the full attention on the whole stack enablement of UALink.

Today, the UALink ecosystem is extremely weak. The spec is there, but there is no solution outside. There is no GPUs on UALink. There is no switches on UALink. If these two combinations are not there, it's just a paper tiger, however good it is. Because of the lack of ecosystem and Broadcom dominating the Ethernet side and coming up with products today and also the hyperscalers who are building their own XPUs, they are more comfortable on the Ethernet side of the stuff because they have the domain expertise and the investments already on the Ethernet side. If you ask my opinion, the scale is a little bit more heavier on the Ethernet side unless and until AMD comes quickly and hits back and says, "Here is a full solution, here is the whole rack," which AMD positions it only to happening in 2027. In the AMD AI Day [sic], the Helios rack and the other aspects of what they showed, today, as an example, it's really abnormality in the industry, the Helios rack is actually running UALink over Ethernet. It is over Ethernet.

That's a little bit of a shame that you position UALink, but you're still deploying it commercially over Ethernet. That gives Ethernet over scale-up a testimony that, "Hey, guys, I'm still there. What are you talking about? The UALink protocol, what you're talking about, today is running on Ethernet." It's basically fabricated on Ethernet fabric. That is something where I think so AMD has lost the shine and they have really not outlasted the Ethernet side of stuff in spite of having such a good solid protocol for UALink. Plus there are hyperscalers who endorse UALink, but they have not come forward with the product line with UALink. There's Meta, there is Amazon who are part of the Consortium. As a matter of fact, even Apple is a member of the UALink Consortium, but nothing has come out of these members.

Analyst:
Do you think that the main thing we need right now is actual UALink switch silicon, whether it comes from Astera Labs or I saw Auradine spun out the Upscale AI? A couple of companies are working on the actual switch silicon element of it. I imagine you can't really have a product out in the market until you have the actual switch chip itself. Is that the main bottleneck present-day, as you think about momentum around UALink?

Specialist:
Yes, I strongly believe that that is the requirement, yes.

Analyst:
Astera has just the higher-lane PCIe switch, the X-Series scale fabric switch, it has the PCIe [peripheral component interconnect express] elements of it, but is lacking the UALink dynamics that will make it more NVLink in nature. Is there a big lift to go, in your mind, from an X-Series to a UALink switch chip? Are you worried that Astera won't be able to actually keep something out of quality?

Specialist:
There are two things. Fundamentally, what Astera has is a Gen 0 of UALink, which is based out of 128G. There are two programmes which Astera Labs is trying to target right now. First one of them, I think so they doing it for one of the hyperscalers, which is the Gen 0 of the UALink. It is basically based out of PCIe flip-based design, and that is a 256-byte FLIT, and it operates at 128G speed of the I/Os, which is PCIe Gen 7 speed. That's number one product line. That will be used only for hyperscalers. It cannot be used as a UALink switch. The second product line, what they're doing is a dedicated UALink switch for the likes of industry, AMD and other customers. That is basically a 64-byte FLIT design. It uses 200G I/Os, and it's basically a complex design. This will be the first time where Astera Labs is jumping from their comfort zone of PCIe to Ethernet because 200G is in the Ethernet domain, and this is the UALink 1.0 spec.

I'm not discounting them, but it is not going to be a smooth ride because whenever you have a leap of 200G or in that domain, there is a lot of learning associated with the design, with the I/O technology associated with it. It will be something to watch out for, but again, Astera Labs and their management are pretty strong enough. They have a good dedicated team of engineering talent. They should be able to pull it off, but as I said, it's not a piece of cake that they will take it because they don't have an actual switch expertise. They were good in the PCIe domain, but now they are moving their domain from PCIe switching to Ethernet switching, which is something unique and new to them. They are entering into the territory of the Broadcoms, the Marvells, the Ciscos who are dominant into the switching space.

Analyst:
Amazon is adopting the X-Series for Trainium3, which is the first programme, which is more like PCIe. I feel like that is well-understood as a Band-Aid solution until you get to the second programme, where you get the full benefits of the Ethernet dynamics within the UALink standard itself. Now that we've got scale-up Ethernet with the ecosystem built around it for Tomahawk Ultra, you could still just go with Nvidia GPUs and NVLink or you could wait around to see what this Intel plus NV Fusion situation will be. Do you think that the X-Series chip has much potential to get any other uptake outside of just Amazon or are most of the broader hyperscalers going to take the wait-and-see approach, and when they get to '27, maybe they'll experiment with NVLink and scale-up Ethernet, but not X-Series PCIe? That will put Astera behind a little from a learning perspective. How do you think about how some of the hyperscalers outside of Amazon might be setting themselves up in preparation for 2027, which I think opens up the optionality of how you can go about constructing your AI clusters?

Specialist:
I think so that X-Series initially will be for Amazon. They have a dominant engagement with Amazon, to be honest. Then leveraging that success, they'll try to penetrate more into the APAC territory with the customers like ByteDance, Alibaba, who are looking into the scale-up using PCIe class of interface because China doesn't have a full access to the 200G technology, to be very honest. They are restricted and limited with the current geopolitical situation. Astera can take advantage of that particular limitation and proliferate their solution with the 128G or the PCIe-based scale-up solution in that market. I think so in the recent call also, they mentioned that they have about 10 customers or something like that who are pipeline for their scale-up technology. I do believe at least seven or eight of them are APAC customers, whereas the bigger hyperscalers will not adopt to that. Because it's very custom and limited, they will directly go to the scale-up Ethernet at 200G or the UALink at 200G.

Analyst:
ByteDance, for example, are partnered with Broadcom from an ASIC perspective. Do you think taking ByteDance, but also some of the other ASIC partners that already have been established from a front-end design perspective, does that become an issue for Astera Labs where if you're working with someone from a front-end silicon design standpoint that they're going to try to push what they'd prefer you to use from a scale-up networking perspective and that would dampen the opportunity for Astera if, say, ByteDance is working with Broadcom, or Google is working with Broadcom or just extending that exact dynamic, does that limit the likelihood that they'll have any interest in Astera's product?

Specialist:
The good part of the Astera product line is if that particular custom ASIC has a PCIe port, a PCIe Gen 7 port, then the Astera solution just is a plug-and-play and they can get the leverage of the scale-up using that architecture. It doesn't require any special ingredients coming from Astera to enable that interface. They just need a standard PCIe Gen 7 port on that design. Irrespective to whether the ASIC is being done by Broadcom or any other ASIC vendor, if the ASIC vendor is smart enough and incorporates a PCIe port

without telling all the details to Broadcom, why and all the stuff, he should be able to comfortably design and use Astera Labs, but on the other side of the spectrum, then he is currently locked and tied to Astera. That gives Astera a lot of leverage that there's no other alternative solutions today, and Astera, it'll be completely a semi-proprietary type of interface because it was the Gen 0 of UALink which nobody adopted, and Astera took it because they had this lead customer who wanted them to build it.

Again, coming back to my previous point, since the UALink ecosystem is so weak, there aren't many players who have shown a commitment to come up with a solution today, except Astera and Auradine and some other vendors. Astera, being the first, gets the advantage of the first-mover advantage. They can lock those accounts and accelerate their path to a very healthy revenue coming from a scale-up solution. Plus they don't have the Ethernet expertise. They will be in a spectrum where there is Ethernet scale-up through Broadcom and others like Marvell and Cisco. Then there will be a PCIe-based scale-up, which is coming from Astera Labs. The Ethernet-based scale-up is not going to dominate in China because of the I/O limitations, which I mentioned to you before. They just do not have the access to that I/O. For them, Astera solution is the most viable solution at a very economical price.

Analyst:
I think it was late last week that everyone knew that Auradine had been working on that, and then they formalised it around this Upscale AI spin-out. They got some funding. There's no strategic funding, which I thought was. Qualcomm Ventures, you could quasi-count as strategic funding, but not any big players backing them. On Upscale AI, is this a distraction or should I continue to pay attention to them in this landscape?

Specialist:
I really don't know, but don't they have funding from Mayfield and all those stuff? I thought they had funding even from Mayfield.

Analyst:
Yes, they do. I meant more like there weren't really any strategic companies that put up some money alongside or at least that wasn't part of the press release.

Specialist:
Honestly, I am puzzled as well because there's too many stuff in that Auradine, then there was AuraLinks, and then now there is Upscale AI, and then within Upscale AI, there are different ventures. I think so Rajiv is the one who is the president of all these companies and orchestrating it. He was the previous CEO of Innovium as well. I wouldn't discount him. He does have the switching expertise and he has taken a successful switch company and incubated it. I don't have much to say. There aren't many players in scale-up, and he is the only one start-up which is showing up. Astera is not a start-up. It's a well-established company. For Upscale to start doing things from scratch, doing the system, doing the software, it's not easy for start-up companies to do switching as a matter of fact. If you notice, there hasn't been a successful start-up which has gone IPO with just a switching technology. You can look at the examples of Barefoot. It was acquired by Intel. Innovium was acquired by Marvell. Overall, anything with the switching start-up stuff, it's a very daunting and a very big task to undertake for a start-up to get switching done and get everything in the stable zone.

Analyst:
If I'm potentially getting the chance to chat with Rajiv, probably not for a handful of weeks, because they've been busy with launching Upscale AI formally. If you had a couple of key questions, what would you be asking?

Specialist:

Key question for them is basically the main part of the switch is the I/O technology. Where are you getting this I/O technology from? Who's enabling you for your I/O technology to be successful? The number two is basically you're dealing with 200G I/Os and you will be having hundreds of them on your design. In terms of the ASIC capability, do you have the right expertise to do such a complex ASIC and get it out? That's number two. Number three is the whole issue of scale-up is memory semantics. It is very intensive in terms of latency and software. How do you manage that challenge that you will be able to succeed and you will be able to get it?

Number four is a very important question that you're putting entire bets on UALink, whereas the market is balanced between Ethernet and UALink. What gives you that confidence that you have gone ahead and invested so much on the UALink camp? Do you have an existing customer or a line of sight where somebody has signed up with you that they're going to deploy your technology? Because UALink is not there in the market today. There is no endpoint. There are no hosts. There are no GPUs. There are no XPUs. It's a big commitment for a start-up to do this without having a handhold or a blessing from somebody who has given you the path that, "Okay, do it and I will buy it from you." Where is that? Can you spot some light on that business aspect of it that who is the lead customer for whom you are driving this for?

Analyst:
For legacy Innovium, I feel like with all these start-ups, there's usually a couple of hyperscalers that really back some of these companies. Is there a hyperscaler that stands out that legacy Innovium had a pretty close relationship with that Rajiv could maybe carry over?

Specialist:
I think so Innovium had a strong relationship with Amazon, but right now, Amazon is in the Astera Labs camp because they've already signed up with them. I really don't know, to be honest.

Analyst:
I was wondering if there was going to be a non-Amazon hyperscaler.

Specialist:
I don't see any non-Amazon guy who signed up with UALink, to be honest. Microsoft is not on the wagon of doing UALink for sure. Google is on OCS, so they don't need UALink. Those two have gone off the back. Meta is basically more on the Ethernet side of scale-up because they are completely a Broadcom house.

Analyst:
How are you thinking about what Nvidia might be liking that they saw from Enfabrica? They didn't take the chip. It wasn't the chip, but there are some interesting things they're doing around. They were both Ethernet and PCIe. That plays well for all the reasons we talked about. There are also some elements around CXL [compute express link], but I don't know if it will be that. Right now, it seems like it's mostly from a scale-out perspective, but I don't know. How are you thinking about what likely drove Nvidia to be interested enough to bring Rochan and his team on board?

Specialist:
The problem with Enfabrica was it was basically a SuperNIC, like a Swiss army. Rochan was trying to do too many things at one time. It came to a point that at the end of the day, they were not having any customer success and it became a huge problem to solve, given the start-up bench strength what they had, but they definitely have a good concept. Nvidia was already an investor in that particular organisation. I'm happy for Rochan that he got the right evaluation and a great exit for him, but on the Nvidia side, what they're trying to do is they have done the memory fabric for the scale-up with the

GPUs, but they want a similar type of environment for the CPUs. Enfabrica brings that kind of a flavour where now they can have a memory fabric for the CPUs and then they have a memory fabric for the GPUs. The GPU memory fabric is through the NVLink. The fabric for the CPUs could be through the SuperNIC, what Enfabrica is designing. This is very intensive software kind of a project where it requires a lot of software building up on the system side, and that's where the Enfabrica lacked the gas to get it done to the final touch-up on their silicon.

With the combination of Enfabrica, Nvidia will be able to explore more on the CPU side with the memory semantics and memory fabric engagement. That will go very well with the combination of ConnectX, which they have it through Mellanox, and Enfabrica's domain expertise. Plus Rochan has a good expertise even on the Ethernet side of switching, having done switching at Broadcom for a long period of time. That combination will help Nvidia to fuel their stuff. Plus there is a lot of FUT coming from Broadcom on the Ethernet scale-up side of stuff. Rochan will be able to handle the Broadcom FUT well enough because he's been seasonal [sic]. He's worked at Broadcom for a good period of time. He knows how the Broadcom think tank works, and he'll be able to complement that at Nvidia to handle all the, I would say, positioning what Broadcom has been putting against Nvidia and in the industry as well.

Analyst:
On the memory fabric for CPU, I know Enfabrica was exploring CXL. Is that what you're expecting that fabric to be based on? I feel like we've been in the CXL hype cycle for probably going on 10 years now, or do you think it will be something different?

Specialist:
It has to be something different because that CXL thing did not fly, to be honest. It was there for a very long period of time, and it really did not take off because CXL did not scale to that level. It was limited to a certain amount of nodes, and there was no software stack. I think so the main problem which lies over here is the software stack where Nvidia can supplement Rochan to get his dreams and get him to the finish line because Nvidia has a super-set of software engineers who have been working on this. They have done Cuda and all those stuff. They will be able to take him to the finish line and get him that vision what he has on the memory fabric for CPUs.

Analyst:
I'll be curious to see what we might see. It won't be Enfabrica-branded, but I feel like we'll see an announcement and know that Rochan and his team might have played a role in it.

Rubin CPX got announced at the AI Infra Summit. It's an interesting take on first token inferencing and eliminating the KV [key-value] cache bottleneck that has existed. I feel like a lot of the hyperscalers' efforts around custom ASICs have been tuned specifically to their own inferencing workloads. Do you think that Rubin CPX answers some of those challenges?

Specialist:
Definitely, the Rubin CPX does address those challenges, especially the HBM capacity has increased significantly in Rubin. Plus Rubin has got more GPU cores inside it. It is taking to the next generation of their current generation GPUs what they have. I would also add one more element to the Rubin thing which a little bit surprises me that it is still in the 3nm technology node. In that way, I think so Jensen and the Nvidia team are still able to extract value from the TSMC 3nm and they didn't have to go to 2nm. That means they will still have a very healthy margin on their business from the Rubin SoC.

Analyst:
We've got the OpenAI announcement with Nvidia, 10GW of capacity, probably [billions of dollars] worth of accelerator and system revenues that might be associated with that. I feel like the industry

outside looking in has been hotly debating which ASIC provider OpenAI might go with. I do think that there are still opportunities for OpenAI to take a bit of a heterogeneous approach because, if you believe Sam Altman, [billions of dollars] is leaving plenty of room for the trillions of dollars he plans to spend. All of it taken together, how do you think about this OpenAI announcement and what it means to the custom ASIC TAM?

Specialist:
If you remember, Broadcom's Hock Tan's update in the recent post. His biggest win is at OpenAI right now. OpenAI ASIC has been awarded to Broadcom, which they awarded last. I would say, later part of last year in Q3, they got that design win. They are doing an accelerator at large scale and Broadcom is their ASIC partner. Addressing your question about ASIC with OpenAI, it's a Broadcom account right now. It's a pretty huge design win for Broadcom, and that's why Hock was claiming the engagement. Of course, a majority of the share is OpenAI. There are a few other design wins he has like the Google TPU and all those stuff, but the biggest mover and shaker is the OpenAI win what they had.

With this engagement of Nvidia, what Nvidia has done is basically taken away some more share of Broadcom format because instead of them investing more on doing custom ASIC and doing all the stuff, Jensen basically has given them a sweet deal and forced their GPUs and given them the priority that, "Don't look outside, don't do anything funky and investing in other things. Just take my stuff, and I will give it to you and solve your problem." He has taken also some more share from OpenAI. Instead of OpenAI going to Broadcom and doing more custom stuff, now they are going to even do Nvidia stuff. I think so going to Nvidia was primarily because of the readiness what Nvidia had to serve them and also they didn't have to put their bets on custom ASIC to such a high level where their deployments get delayed.

Analyst:
I feel like people are most concerned about probably two main things. One is that the custom ASIC TAM shrink is based on all the stuff that we heard over the last month or so, primarily from Nvidia. I think the other one is that NVLink has increased its chances of becoming a de facto standard for the connectivity piece of the AI cluster. As a result, the PCIe ecosystem might be facing some significant pressures moving forward. On those two fronts, is there anything that you feel like I might be missing that would help debunk those concerns, or do you think they're reasonable concerns?

Specialist:
No, those are reasonable concerns, to be honest. This market is so dynamic, right now, I would say the silicon semiconductor segment by itself is on steroids. There are so many things happening in terms of the connectivity, the data centre, the compute segment and the XPU segments, and everybody wants to differentiate themselves from each other. Of course, there's a full dominance of players like Broadcom and Nvidia, but there is still a lot of growth for companies to do custom silicon who want to differentiate themselves from others and the innovation at the pace what is happening, especially on the AI, edge AI side of stuff that I would say we still have a good 2-3 years of healthy run rate on the silicon side of the business. I wouldn't say that the custom silicon TAM is shrinking. It's a little bit of a reactive stuff when people in the market see that the (inaudible) of the AI is still not finalised. People do see sometimes jitters and when they see their earnings not being aligned, they cut back their investments, but then they again come back to track after a few months or after a few quarters. Those things are still there, a little bit of a swing in the pendulum, but the momentum is still there towards semiconductor investment overall, which is fuelling the industry.

Analyst:
Do you think that there is a level of diversity of HPC [high-performance computing] and AI accelerator design work that is pushing to the most leading-edge nodes, or is 10 maybe a misdirection, that not all

the 10 are really going to matter? How are you thinking about the diversity of demand for AI accelerators?

Specialist:
I think so people have not yet still fully gone into the 2nm mode. TSMC themselves are struggling in 2nm in terms of their overall yield and performance, and they are guiding more folks on what they call it as the A14 process node what they have. I haven't seen so much of a success in 2nm, except for the guys like the Apples and the high consumer items which people are buying. Even on the compute side of the segment, a lot of designs today are being executed in 3nm as well. As you said, the Rubin itself is in 3nm because it's not just the process aspect of it, but also the scaling that can these devices being. Can they scale to a commercial deployment range and can you produce them in mass volumes and at proper cost points? I'm a little bit sceptical on 2nm. I haven't smelled success on 2nm in the industry where people are boasting about it and saying that, "Yes, I'll be able to do it." That's something which I wouldn't like to do it. I would say we have to take care of it.

Analyst:
I think ASM International cut its H2 outlook, which I imagine is largely gate-all-around driven, which is a 2nm downtick on demand. I agree with you that it feels like maybe things aren't as great as a comment such as customers moving to 2nm would suggest.

Specialist:
As I said, in terms of the infrastructure, even, for example, high-end switching, when you look at the Tomahawk 6 and all, everything has been done in 3nm, but when it goes to 200Tbit of switching, that 2nm becomes pervasive. When we go to 448G SerDes and high optical connection of 1.6T, you will look at 2nm. I think so 2nm and beyond or lower will be only. You will see some traction of them in 2027. I don't see it in '25. I will not see it in '26. You will see 2027 where 2nm product lines will have some kind of gas in them or steam in them, but other than that, I don't see it.

Analyst:
We're at 224G SerDes [serializer/deserializer] right now. I feel like there's a lot of debate around whether or not optical can overtake things at 448G or 448G SerDes adoption will even exist because of all the work done around optical. How do you feel about where we're at from a pricing and cost, but also a reliability standpoint, on the optical side of things? Is it something beyond 448G that optical really starts to matter?

Specialist:
Beyond 448G, people will start looking at co-packaged optics because at that point, you lose the gas on the electrical side for high speed. Your electrical links will not be able to have that much of crosstalk and noise that it will not be able to sustain the distance. At that point, you'll have to basically go for co-packaged optics. That's what companies like Nvidia, Marvell, Broadcom are gearing up for the future for their CPU technologies. That's why you see a lot of photonics companies and photonic start-ups in that space because we are reaching to the end of the tunnel where you cannot scale beyond. That's happening on the 448G. 224G will have a very long life. I would say it'll sustain until 2028 and beyond. It'll have a good 4-5 years of runway. Right now, the designs are being implemented in 224G. Tomahawk 6 is a 224G switching device. Once you have the switching device in 224G, then you will see other peripherals like the SmartNIC cards and the host interface which will support 224G as well. That's what is happening.

448G per se, it is a different beast compared to the 112G and 200G because it has a different PAM technology for electrical and a different PAM technology for optics. In all other cases like 112G and 224G, the PAM technology was common between electrical and optical. You had the same PAM. They would talk to each other, but in 448G, the PAM technology is different for optics and it's different for electrical. That creates a barrier between electrical and optical interconnects and you need a gearbox or a matching DSP to do it. Companies have demonstrated 448G today. I would expect that 448G devices would be there in the market by 2028 timeframe, where you look at the 200T or the 400T switch, but beyond 448G, it's going to be CPU.

Analyst:
Hock Tan commented that Ethernet-based active electrical cables are only in existence because customers still use lower-quality SerDes. If you use high-quality SerDes that AECs [active electrical cables] are not necessary. How do you think about the durability of the AEC market, considering that dynamic?

Specialist:
I think so I don't fully bind to that argument of his because his SerDes might be good because he makes the switch, but people who are doing GPUs and doing NIC cards and all, they don't need to put such a strong SerDes inside and everybody is not going to put a strong SerDes. He's talking only for himself. He should be looking at everybody is not coming from Broadcom, everybody is not using Broadcom devices. From an economies of scale, you look at the whole ecosystem, AECs will exist because there will be Broadcom and there will be non-Broadcom. When non-Broadcom wants to talk to Broadcom, you will use AEC. Everybody cannot be using Broadcom. He's just trying to brute force that if you use me, then you don't need AECs, but then you're paying the cost of going through Broadcom and paying them a heavy amount of dollars. In that sense, I would say the AEC market will still exist. There are companies like Credo, Marvell, Astera Labs who are using it. There are hyperscalers who are doing the custom silicon like the Amazons and the Googles, and they will use AECs. From an ecosystem point of view, AECs do exist today, and it will exist in the future as well. It's not going to die tomorrow, to be honest.

Analyst:
I feel like Astera has had a tough go from a content story perspective with anything Nvidia-related, and it feels like this Intel custom CPU with an Nvidia GPU is further deflationary for both their retimer content as well as that head node switch chip that they have for P-Series, because customers might just customise less. Do you think that the Intel-Nvidia partnership is a further negative for Astera and beyond just the X-Series part of it?

Specialist:
Definitely. By the way, Nvidia removed them. After PCIe Gen 5, they didn't need them for Gen 6. The Grace Hopper stuff, first generation used Astera as a retimer for Gen 5, but in Gen 6, they completely eliminated them, exactly the same theory like Hock Tan. If my SerDes are good, I don't need you. They also eliminated the PCIe switch because the ConnectX series now has integrated PCIe switch inside. Nvidia actually eliminated Astera. After their first generation, they never used Astera. They took them off because they realised that they're paying a lot of money to them, so they don't need it and their designs were completely off. With this Intel stuff and infusion of Nvidia inside Intel, they will also remove the PCIe stuff from Intel because everything will be based on the GPU and the NVLink stuff. PCIe dominance will go away from the Intel camp. If the PCIe dominance goes away, then Astera will struggle to have their stuff in Intel Inside.

Analyst:
What do you think about their trajectory, Astera Labs, from here? It seems like it got a lot more challenging.

Specialist:
I think so they will proliferate into the switching business of Ethernet and the retimers. They have a lot

of capital in terms of stock value. If I was in their case, I would do some strong M&A with some companies and grow their footprint in different areas. They just cannot be vertically integrated with the limited stuff what they're doing in the PCIe and the UALink space. They should spread their wings right now, given their bench strength what they have, and leverage their stock values to buy some companies or buy some start-ups so that they can grow their footprint. They should start proliferating into optics, DSP in those type of spaces.

Analyst:
Do you think the optics and DSP [digital signal processing] market is just much more competitive? Do you think it's likely that they replicate their success they saw in retimers in any product category in the future?

Specialist:
I think so that the DSP market is pretty healthy and strong because it is being deployed on the optics side. It is a market which is quite evolving from 800G, 1.6T, 3.2T. There is still a lot of fuel in that market to grow. If you have the right technology, you can do wonders. You look at MaxLinear. They used to be a USD 9 stock just six months back. Today, they are USD 17 stock. It's given you 100% returns in six months. Astera should buy a company like MaxLinear as an example. They can easily acquire that company and leverage it and do something much better than what MaxLinear is doing today.

Analyst:
I feel like that will be reacted to pretty negatively because I think everyone has a legacy interpretation of what kind of company MaxLinear is.

Specialist:
I understand. I get it, what type of a company it is, but they can restructure it because they can infuse their thinking, and they buy for talent and their technology and then benefit from that price.

Analyst:
Talking about the end of last year, everyone thought 2025 would be the peak growth year of AI. Towards the end of this year, I feel like people are thinking '26 is going to be the growth peak of AI. How are you thinking about the durability of AI demand, especially considering some of the big announcements we've seen across a lot of the hyperscalers lately?

Specialist:
I think so the AI still has a lot of legs to run, especially it is proliferating in the edge side of the component side, on the inference side of things. People are basically trying to look at more optimised way of doing the AI models right now. A lot of, I would say, R&D and research is going on in doing parallelism on these models which are there and especially on the things called as the physical AI, which is coming into play, a third dimension to AI, I would say. A lot of emphasis today has been put on the physical AI side of stuff, which will make it much more accurate and much more human. With that, the compute power is still going to scale further ahead to get that reality into the AI. It is still going to get fuelled.

Companies might say whatever they want, but ultimately, the industry will drive towards that, and they will be forced to do that investment to get to the next stage. Otherwise, they'll be out of the race. How much ever Microsoft tells about cutting down the investments and all the stuff, but you look at Meta, you look at Amazon, they are going double down on AI right now. Especially Meta is investing like crazy in terms of their talent pool, in terms of their infrastructure. They are just going all out on the AI side of stuff. They will reap the benefit, if not today, then tomorrow. If they are not investing, then they will be out of the race.

Analyst:
If you think about a specific technology where the industry is still debating on what the right approach is going to be, it's CPO [co-packaged optics], scaled networking. Thinking about liquid cooling introduction, just if you picked a technology that you think is most debated that's going to have the biggest impact on what the trajectory of the landscape ends up being, what would you say I should isolate down to one or two of those to really dig in to try to understand the moving pieces and try to formulate a view on where the industry actually goes?

Specialist:
I would say two things. One is the co-packaged optics. A lot of things are happening there. Just as a reference, the main challenge in co-packaged optics today is that it's not being manufactured in mainstream fab. It's basically being done in gallium arsenide. That is the main stuff what is there. That makes a difference to where we are on the co-package optics because when you're doing in gallium arsenide, you're not in the mainstream TSMC and then you don't get the economies of scale, those fabs are not manufacturing in thousands and millions of those ports. That is a bottleneck. The second aspect of it is different technologies are existing in CPU by itself in terms of MGM, EMI [sic] and all those type of things or EMR or ring-based photonics. Those are the elements which the industry has to converge. Then there is another aspect of serviceability and manageability of these co-packaged optics, the fibre and all the stuff, the ecosystem. A lot of stuff has happened in the past few years and a lot of stuff is going to happen in the coming few years to get this technology stable because this is the only technology which will scale going forward, and you cannot have the pluggable optics in the future. For example, how we went from the cassette player to the CD and then to the MP3 digital audio, similarly, this optic revolution is going to happen.

We had huge optical connectors before like a cigarette pack. Then we went to these modules which are being used today. Then the next generation is going to be co-packaged. It's imminent, and that's where the majority of discussions, investments are happening today. That's one part of the equation. The second part of the equation is there will be two networks existing in the data centre. One will be for the compute and one will be for the network. These two networks will exist. The growth today is not on the network side. It's a myth that the network is going to grow. In reality, what is going to grow is the compute side of the network, where you're connecting all the CPUs and the GPUs because that's where the real horsepower is. The connectivity of network already exists today. You can still leverage what the investments we have been doing since the past 10 years on the scale-out. You don't really need to put your energy into that today. It's there. Where the entire industry energy is going to be right now is networking of the CPUs and the GPUs, which didn't exist before because those are the highest OPEX of the network. Remember that, the highest OPEX what you're spending today is on the GPU and CPU. You want to make sure you're utilising them to the best and leveraging them to the best.

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